

A rural landscape featuring a large electricity pylon in the background, with banana trees and a field in the foreground. Two people are visible in the field, and a cow is lying down. The sky is overcast.

Evaluating Rural Electricity Access

Content:

1. Theoretical Background and Motivation
2. Choice of Variables
3. Models and Estimations
4. OLS Assumption Tests
5. Results and Limitations

Motivation

- Access to Electricity remains a key feature of sustainable development
- Increases basic living standards, economic participation and opportunities
- Past experiences show relevance of government policies in providing rural electricity
- Research suggests link between policy effectiveness and energy access

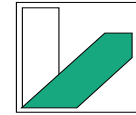
Data Source

- World Bank Databank ¹
- 2016 data from World Development Indicators

Methodology

- Bivariate and Multivariate Regression Models in R with OLS Assumption Tests

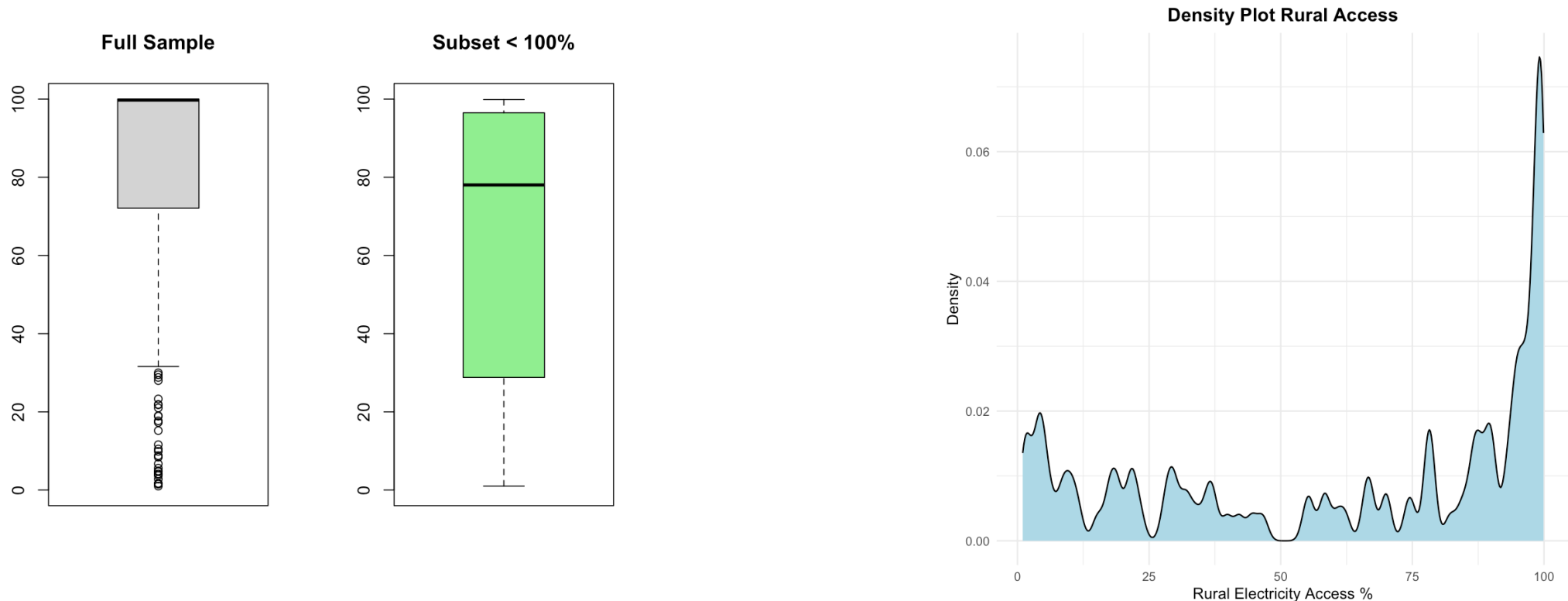
¹ <https://databank.worldbank.org/source/world-development-indicators>



Is rural electricity access determined by government effectiveness?

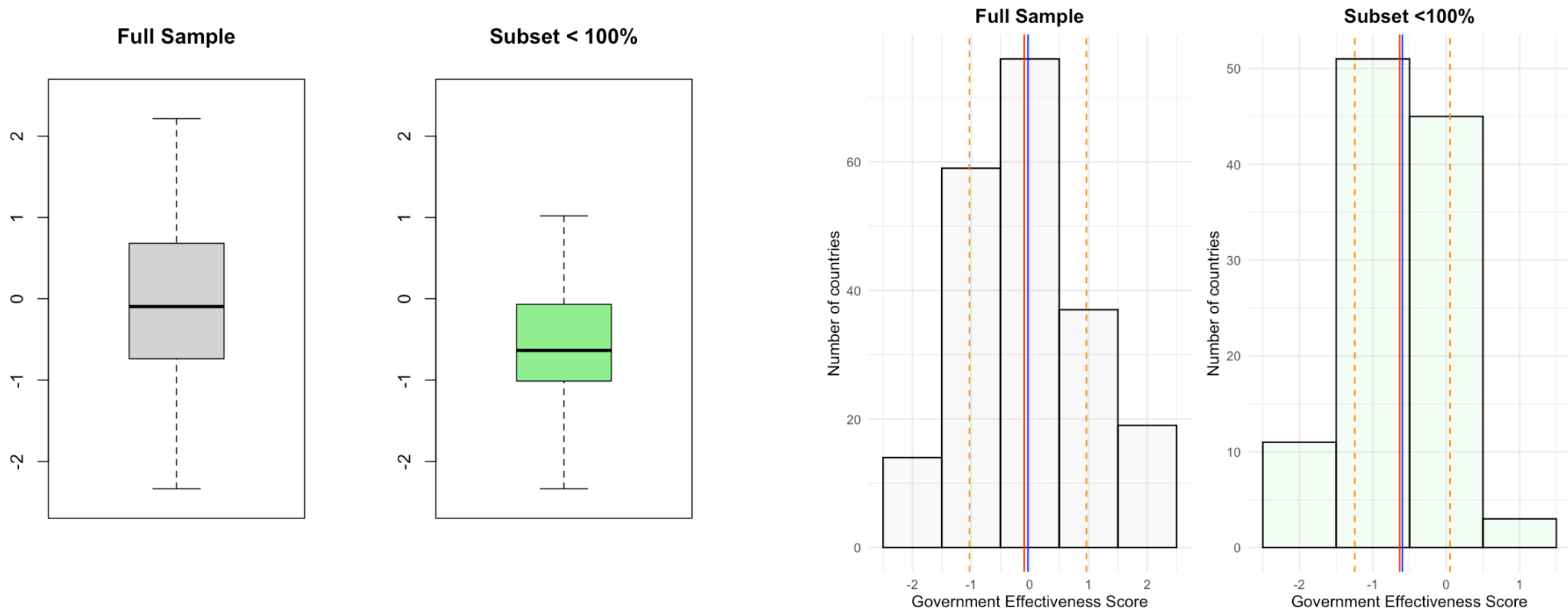
Dependent Variable – Rural Access to Electricity

- Describes the amount (in %) of rural population with access to electricity
- **Choice to subset data for countries below 100% rural electricity access**
- ➔ At 100% no further increase in electricity access possible, irrespective of change in government efficiency



Independent Variable – Government Effectiveness

- Government Effectiveness measures the quality and implementation of policies and interventions, credibility of government decisions, and effectiveness of government services.
- Measured on a scale from -2.5 to +2.5, values derived from questionnaires on perception of experts, NGO and policymaker opinions.



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Dependent variable:

Rural Electricity Access %

gov_eff 31.265
 (4.414)
 p = 0.000***

Constant 81.798
 (3.884)
 p = 0.000***

Observations 110
R2 0.317
Adjusted R2 0.311
Residual Std. Error 29.896 (df = 108)
F Statistic 50.180*** (df = 1; 108)

Note: *p<0.1; **p<0.05; ***p<0.01

Interpretation

- Government effectiveness has a highly significant (***) effect on rural electrification.
- *An increase in government effectiveness by 0.1 points on the scale, leads to an increase of rural electrification by approx. 3.127 %, ceteris paribus.*

Evaluation

- Model explains approx. 31% of the variation of DV.
- (Adjusted) R2 are reasonable for bivariate model, standard error of IV is not concerning
- N=110 is sufficient for interpretation

Grid Loss

Reasoning

- Grid Loss describes the quality of the energy grid. It measures transmission and distribution losses between sources of supply and points of distribution caused by faulty infrastructure and inefficiencies in the grid. Reasoning is that with more inefficient grids, economic feasibility of grid expansion might be lower.
- ***Maintenance and improvement of already existing grids might draw investments and allocated budgets - rather than grid expansion to rural areas.***

```
> summary(dta_low$grid_loss)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  4.582 10.086 14.738 17.170 20.747 58.184     30
```

(log) GDP per capita, PPP²

Reasoning

- GDP per capita is a measure of economic activity, wealth, and prosperity of a given state. Grid expansion is costly and requires financial capabilities of a government. Citizens must be able to afford electricity and appliances that require electricity.
- ***Economic activity is necessary for large scale, costly infrastructure investments. Absence of economic activity and lower expected returns hinder large scale infrastructure investments.***

```
> summary(dta_low$logGDP)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  6.639  8.034  8.704  8.689  9.418 10.318     6
```

3 Multivariate Model Comparison

Dependent Variable:			
Rural Electrification (%)			
	(1)	(2)	(3)
Government Eff.	31.265*** (4.414)	26.363*** (5.514)	9.988* (5.768)
Grid Losses		-0.393 (0.329)	
log GDP per Capita			24.787*** (3.983)
Constant	81.798*** (3.884)	88.676*** (6.444)	-146.624*** (36.947)
Observations	110	80	104
R2	0.317	0.281	0.523
Adjusted R2	0.311	0.262	0.514
Residual Std. Error	29.896 (df = 108)	29.446 (df = 77)	25.167 (df = 101)
F Statistic	50.180*** (df = 1; 108)	15.041*** (df = 2; 77)	55.445*** (df = 2; 101)

Note: *p<0.1; **p<0.05; ***p<0.01

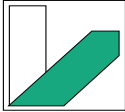
Interpretation

- Significance and Effect of Government Effectiveness reduce when controls are included.
- Effect only significant at 10% level.
- LogGDP has the most significant effect on electricity access in rural areas ($p < 0.01$).
- An increase in government effectiveness by 0.1 points, leads to an increase in rural electricity access of 1%, ceteris paribus.*

Evaluation

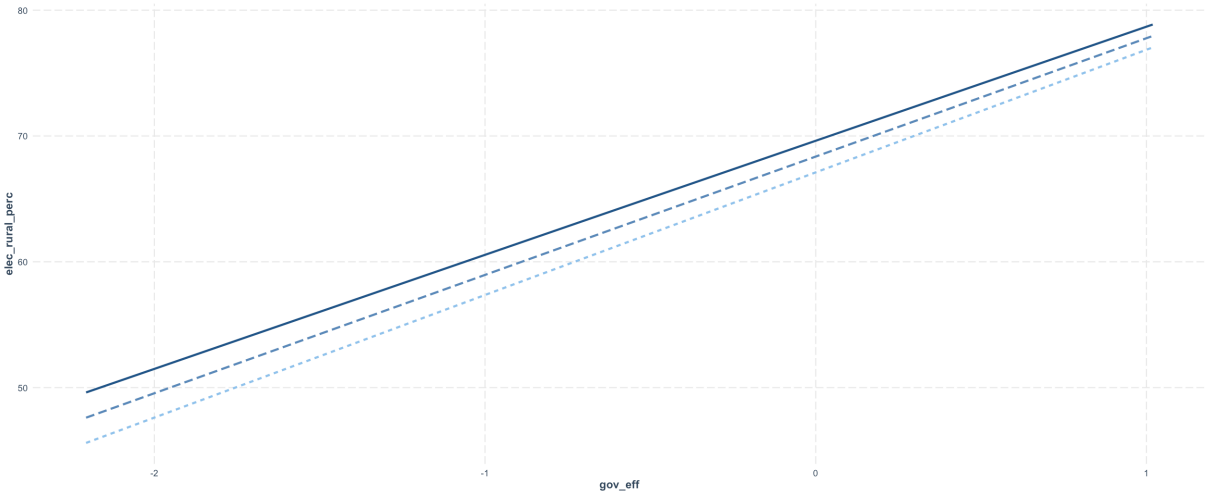
- Government effectiveness remains marginally significant once GDP per capita is controlled for.

3 Introducing the Interaction Term



Interaction Term

- Using (*government effectiveness * rural population*)
- With higher rural population shares, the task of electrifying rural areas may become larger and more costly.
- Interaction term captures the difficulty of the task.
How relevant is government effectiveness given the size of rural population in a country?



Dependent variable:

elec_rural_perc

gov_eff	10.257 (14.051) p = 0.468
logGDP	26.104 (5.061) p = 0.00001***
rural_pop_share	0.062 (0.189) p = 0.744
gov_eff:rural_pop_share	-0.017 (0.241) p = 0.945
Constant	-161.583 (51.525) p = 0.003***

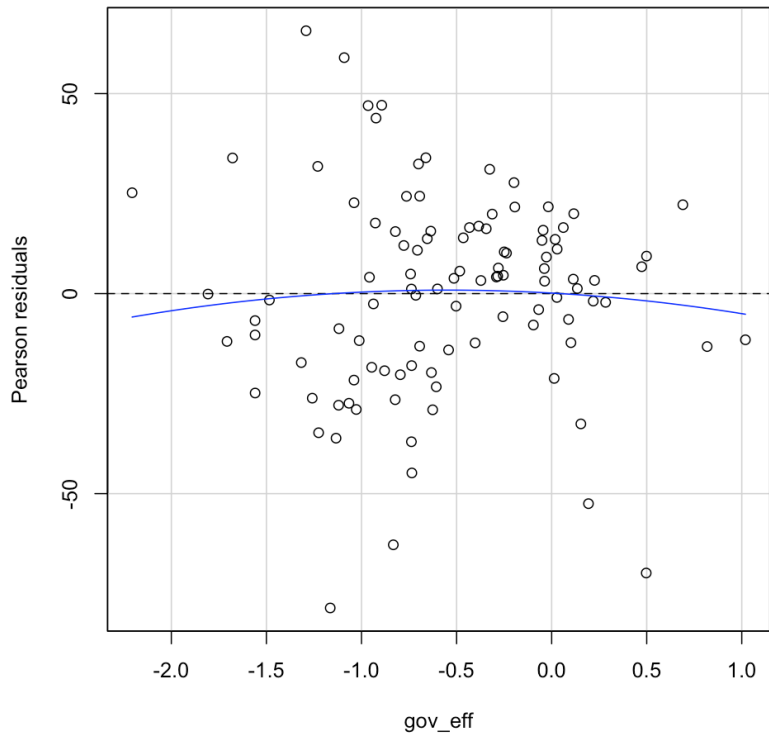
Observations	104
R2	0.524
Adjusted R2	0.505
Residual Std. Error	25.396 (df = 99)
F Statistic	27.270*** (df = 4; 99)

Note: *p<0.1; **p<0.05; ***p<0.01

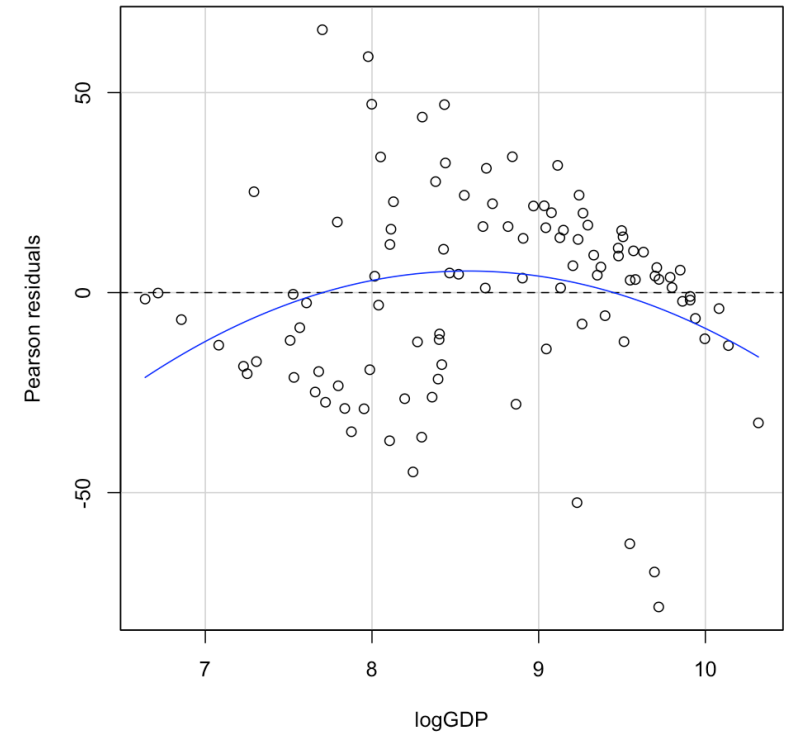
OLS Assumptions – Zero Conditional Mean

- Using *residual plots*, we can test for zero conditional means assumption. The blue line should ideally be **horizontal** across the plot, signaling **zero conditional mean**.

Residual Plot for Government Efficiency



Residual Plot for logGDP per Capita



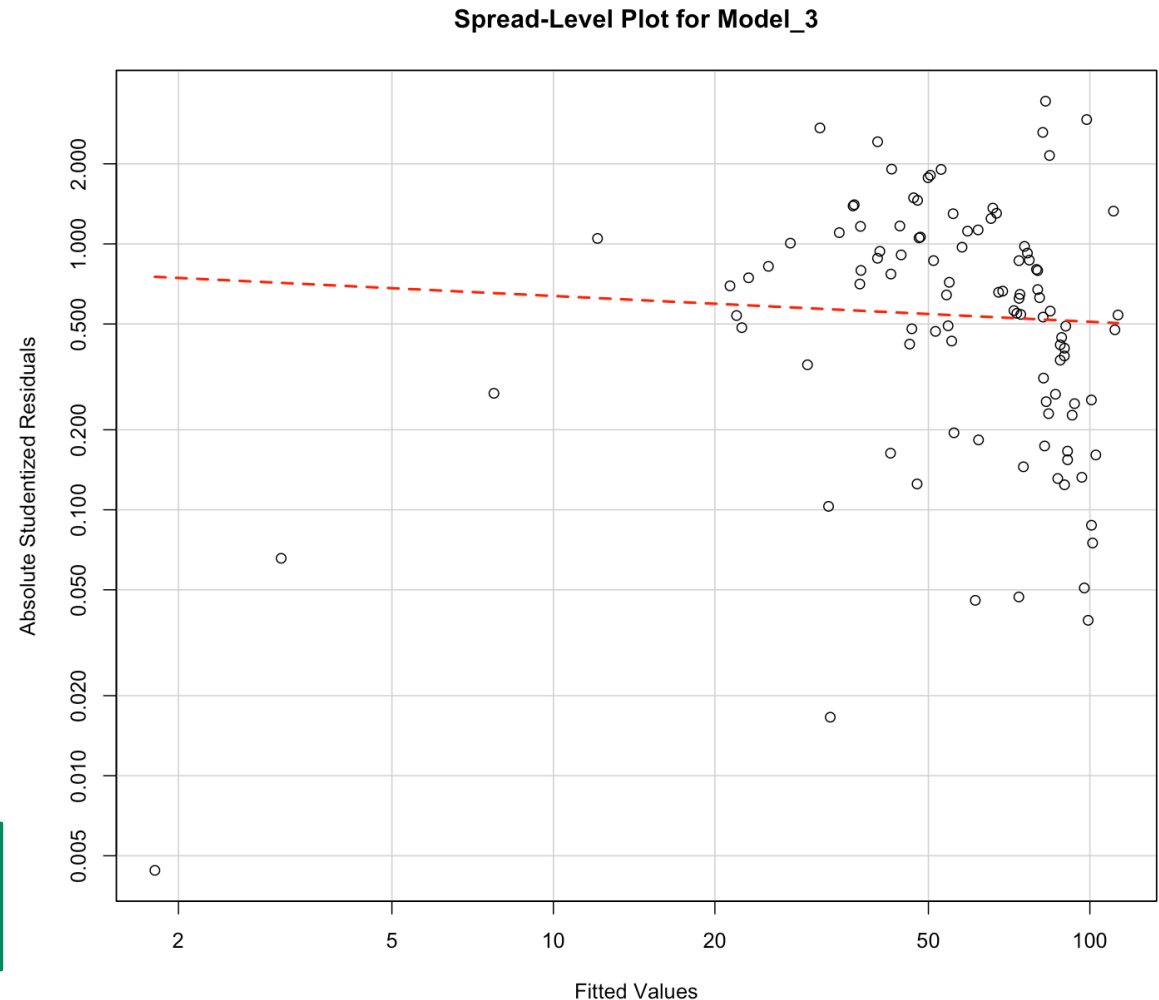
OLS Assumptions - Homoscedasticity

- Using *spread level plots* in R, we can check for Homoscedasticity in the model.
- Horizontal Red Line: **Homoscedasticity**

Interpretation

- Heteroscedasticity **not** overly concerning
- OLS Assumption holds just considering this test.

Should not be overreadily accepted without further testing.



Results

- Government efficiency, including policy quality and implementation and effectiveness of government services **does** have a significant effect on rural access to electricity. An increase in government effectiveness, *ceteris paribus*, leads to an increase in rural electricity access.
- However, rural electricity access, according to this model and underlying data, is best explained through *GDP per capita, adjusted for purchasing power*.
- **The relation and direction is coherent with other findings, suggesting a relationship between infrastructure investments and GDP per capita.**
- Government effectiveness remains a relevant, significant variable for explaining rural electricity access.

Limitations

- **Interpretability of IV** - Unclear how a 0.1 change in Government Effectiveness can be achieved
- **Old data** - *policy relevance not as strong*
- No independent variable that captures rural population density or geographical challenges sufficiently
- Possible **multicollinearity**:
 - ➔ Wealth of nations is determined by the quality of their institutions (Acemoglu & Robinson, 2012)³
 - ➔ GDP and government effectiveness might correlate over time, causing multicollinearity
- **Further Panel-Analysis is required**

³ Acemoglu, D., & Robinson, J. A. (2012). *Why Nations Fail: The origins of power, prosperity, and poverty*. Profile Books.